Beamforming Optimization for Cell-Free ISAC Systems: Mathematical Derivation and Solution Methods

Research Stage Report

June 17, 2026

Abstract

This paper establishes a mathematical framework for joint beamforming optimization in Cell-Free Integrated Sensing and Communication (ISAC) systems. Considering communication quality-of-service (QoS) constraints, sensing detection probability constraints, tracking accuracy constraints (PCRB), and access point (AP) selection constraints, we derive the semidefinite programming (SDP) relaxation formulation of the robust optimization problem under imperfect channel state information (CSI). Through the S-Procedure, infinite-dimensional worst-case constraints are transformed into finite-dimensional linear matrix inequalities (LMI). The convexity of sensing constraints under the single-angle estimation assumption is rigorously proved, and rank-one recovery algorithms along with closed-form solution schemes are provided. This document serves as the theoretical foundation for subsequent algorithm implementation and paper writing.

Contents

1 Problem Formulation

1.1 Optimization Problem (Standard Form)

Consider a Cell-Free ISAC system with M distributed APs, K communication users, and P sensing targets. Each AP is equipped with N_t transmit antennas. The optimization objective is to minimize the total transmit power while satisfying communication QoS, sensing detection, tracking accuracy, and AP selection constraints.

$$\min_{\substack{\{\mathbf{w}_{m,k}\},\{\mathbf{Z}_m\},\\ \{b_{mp}\}}} \quad \sum_{m=1}^{M}\left(\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m)\right) \tag{P1}$$

$$\text{s.t.} \quad \mathrm{SINR}_k^{\mathrm{wc}} \geq \gamma_k, \quad \forall k \in \mathcal{K} \tag{1}$$

$$\mathrm{SINR}_{S,p} \geq \gamma_S^{\mathrm{PoD}}, \quad \forall p \in \mathcal{P} \tag{2}$$

$$\mathrm{tr}(\mathbf{J}_p^{\mathrm{data}}) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P} \tag{3}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{4}$$

$$\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|_2^2 + \mathrm{tr}(\mathbf{Z}_m) \leq P_{\max}, \quad \forall m \in \mathcal{M} \tag{5}$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \tag{6}$$

$$b_{mp} \in \{0,1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \tag{7}$$

where $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$ denotes the communication beamforming vector from AP m to user k, $\mathbf{Z}_m \in \mathbb{C}^{N_t \times N_t}$ represents the sensing covariance matrix at AP m, and $b_{mp} \in \{0,1\}$ is the binary indicator variable for whether AP m serves target p. The parameters $P_{\max}$, γ_k, γ_S^{PoD}, $\Gamma_{\mathrm{Track},p}$, and N_{req} denote the per-AP maximum transmit power, the SINR threshold for user k, the sensing detection probability threshold, the tracking accuracy threshold (FIM trace constraint) for target p, and the required number of serving APs per target, respectively.

1.2 Notation Table

Table 1 summarizes the main notation used throughout this paper.

Remark 1. *Specific numerical parameters (e.g., $M = 16$, $N_t = 4$, $P_{\max} = 30$ W) will be provided in the Simulation Setup section. The theoretical derivations herein maintain general symbolic notation.*

2 System Model and Signal Model

2.1 Communication Signal Model

The received signal at user k can be expressed as:

$$y_k = \underbrace{\sum_{m=1}^{M}\mathbf{h}_{m,k}^H\mathbf{w}_{m,k}s_k}_{\text{desired signal}} + \underbrace{\sum_{j\neq k}\sum_{m=1}^{M}\mathbf{h}_{m,k}^H\mathbf{w}_{m,j}s_j}_{\text{multi-user interference}} + n_k \tag{8}$$

Table 1: Main Notation Table

Symbol	Dimension	Physical Meaning
M	Scalar	Total number of APs
N_t	Scalar	Number of transmit antennas per AP
K	Scalar	Total number of communication users
P	Scalar	Total number of sensing targets
$P_{\max}$	Scalar	Maximum transmit power per AP
γ_k	Scalar	SINR threshold for user k
γ_S^{PoD}	Scalar	Sensing detection probability threshold
$\Gamma_{\mathrm{Track},p}$	Scalar	Tracking accuracy threshold for target p
σ_c^2	Scalar	Communication receiver noise variance
σ_s^2	Scalar	Sensing receiver noise variance
ϵ_h	Scalar	Relative error bound for communication channel estimation
ϵ_g	Scalar	Relative error bound for sensing channel estimation
N_{req}	Scalar	Required number of serving APs per target
$\mathbf{h}_{m,k}$	$\mathbb{C}^{N_t}$	Communication channel from AP m to user k
$\mathbf{g}_{m,p}$	$\mathbb{C}^{N_t}$	Sensing channel from AP m to target p
$\mathbf{w}_{m,k}$	$\mathbb{C}^{N_t}$	Communication beam from AP m to user k
$\mathbf{Z}_m$	$\mathbb{C}^{N_t \times N_t}$	Sensing covariance matrix at AP m
b_{mp}	$\{0,1\}$	Whether AP m serves target p

where $n_k \sim \mathcal{CN}(0, \sigma_c^2)$ denotes the receiver noise.

To facilitate analysis, we introduce stacked vector forms. Define the global communication channel $\mathbf{h}_k \in \mathbb{C}^{MN_t}$ and global communication beam $\mathbf{w}_k \in \mathbb{C}^{MN_t}$ as:

$$\mathbf{h}_k = [\mathbf{h}_{1,k}^T, \mathbf{h}_{2,k}^T, \ldots, \mathbf{h}_{M,k}^T]^T, \quad \mathbf{w}_k = [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \ldots, \mathbf{w}_{M,k}^T]^T \tag{9}$$

The received signal can then be compactly written as:

$$y_k = \mathbf{h}_k^H \mathbf{w}_k s_k + \sum_{j \neq k} \mathbf{h}_k^H \mathbf{w}_j s_j + n_k \tag{10}$$

2.2 Sensing Signal Model

For monostatic radar sensing, the reflected signal from target p is:

$$y_{S,p} = \sum_{m=1}^{M} \mathbf{g}_{m,p}^H \mathbf{Z}_m \mathbf{g}_{m,p} + n_{S,p} \tag{11}$$

where $n_{S,p} \sim \mathcal{CN}(0, \sigma_s^2)$ denotes the sensing receiver noise.

2.3 Imperfect CSI Model

In practical systems, channel estimation suffers from errors. This paper adopts the bounded error model:

Communication channel:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \Delta\mathbf{h}_k, \quad \|\Delta\mathbf{h}_k\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}_k\|_2 \tag{12}$$

Sensing channel:

$$\mathbf{g}_p = \hat{\mathbf{g}}_p + \Delta\mathbf{g}_p, \quad \|\Delta\mathbf{g}_p\|_2 \leq \epsilon_g \|\hat{\mathbf{g}}_p\|_2 \tag{13}$$

where $\hat{\mathbf{h}}_k$ and $\hat{\mathbf{g}}_p$ are the estimated channels, $\Delta\mathbf{h}_k$ and $\Delta\mathbf{g}_p$ are the estimation errors, and ϵ_h and ϵ_g are the relative error bounds.

3 Communication Constraint Derivation

3.1 SINR Definition

The signal-to-interference-plus-noise ratio (SINR) for communication user k is defined as:

$$\text{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \sigma_c^2} \tag{1}$$

3.2 Worst-Case SINR Analysis

Under imperfect CSI, we need to guarantee that the worst-case SINR meets the threshold requirement. Considering the worst cases for both the numerator and interference:

Worst-case numerator: When the error direction opposes the beam, the effective channel gain is minimized:

$$|\mathbf{h}_k^H \mathbf{w}_k|^2 \approx |\hat{\mathbf{h}}_k^H \mathbf{w}_k|^2 (1 - \epsilon_h)^2 \tag{14}$$

Worst-case interference: When the error direction enhances interference:

$$|\mathbf{h}_k^H \mathbf{w}_j|^2 \approx |\hat{\mathbf{h}}_k^H \mathbf{w}_j|^2 (1 + \epsilon_h)^2 \tag{15}$$

Therefore, the worst-case SINR can be approximated as:

$$\text{SINR}_k^{\text{wc}} \approx \text{SINR}_k^{\text{nom}} \cdot \frac{(1 - \epsilon_h)^2}{(1 + \epsilon_h)^2} \tag{2}$$

3.3 Robustness Factor and Threshold

Define the communication robustness factor:

$$\eta_h = \left(\frac{1 - \epsilon_h}{1 + \epsilon_h} \right)^2 \tag{3}$$

For $\epsilon_h = 0.10$ (10% error), $\eta_h \approx 0.669$ (-1.75 dB). This implies that CSI errors cause an effective SINR degradation of approximately 1.75 dB.

To ensure the worst-case SINR meets the threshold, the nominal SINR requirement must be raised:

$$\gamma_k^{\text{robust}} = \gamma_k \cdot \frac{(1 + \epsilon_h)^2}{(1 - \epsilon_h)^2} = \frac{\gamma_k}{\eta_h} \tag{4}$$

For $\gamma_k = 1$ (0 dB) and $\epsilon_h = 0.10$: $\gamma_k^{\text{robust}} \approx 1.493$ (1.74 dB).

4 Sensing Constraint Derivation

4.1 Sensing SINR

The sensing SINR for target p is:

$$\mathrm{SINR}_{S,p} = \frac{|\mathbf{g}_p^H \mathbf{z}_p|^2}{\sigma_s^2 \|\mathbf{z}_p\|_2^2} \tag{5}$$

By the Cauchy-Schwarz inequality, the optimal sensing beam is the matched filter:

$$\mathbf{z}_p^* = \sqrt{P_{S,p}} \frac{\mathbf{g}_p}{\|\mathbf{g}_p\|_2} \tag{6}$$

yielding the optimal SINR:

$$\mathrm{SINR}_{S,p}^* = \frac{P_{S,p}\|\mathbf{g}_p\|_2^2}{\sigma_s^2} \tag{7}$$

To meet the detection threshold γ_S^{PoD}, the minimum sensing power is:

$$P_{S,p}^{\min} = \frac{\gamma_S^{\mathrm{PoD}} \sigma_s^2}{\|\mathbf{g}_p\|_2^2} \tag{8}$$

4.2 Robust Sensing Constraint

Similarly, define the sensing robustness factor:

$$\eta_g = \left(\frac{1-\epsilon_g}{1+\epsilon_g}\right)^2 \tag{9}$$

For $\epsilon_g = 0.15$ (15% error), $\eta_g \approx 0.546$ (-2.63 dB).
The robust sensing threshold:

$$(\gamma_S^{\mathrm{PoD}})^{\mathrm{robust}} = \gamma_S^{\mathrm{PoD}} \cdot \frac{(1+\epsilon_g)^2}{(1-\epsilon_g)^2} = \frac{\gamma_S^{\mathrm{PoD}}}{\eta_g} \tag{10}$$

4.3 PCRB Constraint and Fisher Information Matrix

The tracking accuracy constraint is characterized through the trace of the Fisher Information Matrix (FIM). The data-dependent FIM is:

$$\mathbf{J}_p^{\mathrm{data}} = \frac{2}{\sigma_s^2} \mathrm{Re}\left\{\nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p^H \cdot \mathbf{R}_X \cdot \nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p\right\} \tag{11}$$

where $\mathbf{R}_X = \sum_k \mathbf{w}_k \mathbf{w}_k^H + \sum_p \mathbf{z}_p \mathbf{z}_p^H = \sum_k \mathbf{W}_k + \mathbf{Z}$ is the transmit covariance matrix and $\boldsymbol{\theta}_p$ denotes the target state parameters.

Assumption 1 (Sensing Parameter Estimation Model)**.** *This paper considers target state parameters $\boldsymbol{\theta}_p = [\theta_p]$ (single-angle estimation). Under this setting, the gradient matrix $\nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p$ is determined by the target prediction state within the current time slot and can be treated as a known constant matrix. Consequently, $\mathbf{J}_p^{data}$ is an* ***affine function*** *of $\mathbf{R}_X$, and its trace constraint can be precisely formulated as:*

$$tr(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{Track,p} \tag{16}$$

where $\mathbf{F}_p = \frac{2}{\sigma_s^2} Re\left\{\nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p \nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p^H\right\}$ is a known constant positive semidefinite matrix.

Remark 2. *Since this work focuses on direction-of-arrival (DOA) estimation for targets, the relative delay and Doppler shift between the probing signal and the target are regarded as slowly varying parameters within a short observation frame. Under this condition, the FIM degenerates into a constant matrix* $\mathbf{F}_p$ *that depends solely on the angle gradient, thereby guaranteeing the convex affine property of the constraint.*

Proof of FIM Linearity: Let $\mathbf{G}_p = \nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \in \mathbb{C}^{D\times MN_t}$, then:

$$\mathbf{J}_p^{\text{data}} = \frac{2}{\sigma_s^2}\text{Re}\{\mathbf{G}_p\mathbf{R}_X\mathbf{G}_p^H\} \tag{17}$$

Expanding the (a, b)-th element:

$$(\mathbf{J}_p^{\text{data}})_{ab} = \frac{2}{\sigma_s^2}\text{Re}\Big\{\sum_{i,j}(\mathbf{G}_p)_{ai}(\mathbf{R}_X)_{ij}(\mathbf{G}_p^H)_{jb}\Big\} \tag{18}$$

This is a linear combination of the elements of $\mathbf{R}_X$. Therefore, the trace constraint:

$$\text{tr}(\mathbf{J}_p^{\text{data}}) = \text{tr}(\mathbf{F}_p\mathbf{R}_X) \tag{12}$$

where:

$$\mathbf{F}_p = \frac{2}{\sigma_s^2}\text{Re}\Big\{\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\Big\} \tag{13}$$

is a constant positive semidefinite matrix independent of the optimization variables. □

5 SDP Relaxation and Global Variable Reformulation

5.1 From Beams to Covariances

To transform the non-convex beamforming problem into a convex optimization problem, we introduce covariance matrix variables:

Communication covariance matrix:

$$\mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H \in \mathbb{C}^{MN_t\times MN_t} \tag{15}$$

Sensing covariance matrix:

$$\mathbf{Z} = \sum_p \mathbf{z}_p\mathbf{z}_p^H \in \mathbb{C}^{MN_t\times MN_t} \tag{16}$$

Global transmit covariance:

$$\mathbf{R}_X = \sum_{k=1}^{K}\mathbf{W}_k + \mathbf{Z} \tag{17}$$

5.2 SDR Relaxation and Final Convex SDP Problem

The original problem requires $\text{rank}(\mathbf{W}_k) = 1$ (since $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$). Semidefinite relaxation (SDR) discards the rank-one constraint, requiring only $\mathbf{W}_k \succeq \mathbf{0}$. Furthermore, through the S-Procedure, the infinite-dimensional worst-case SINR constraint is exactly transformed into a finite-dimensional linear matrix inequality (LMI), yielding the following **standard convex semidefinite programming problem** that can be directly fed into CVX/MOSEK.

Core Auxiliary Definition:

$$\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j \tag{19}$$

Final Convex SDP Problem (P3):

$$\min_{\{\mathbf{W}_k\}, \mathbf{Z}, \{\mu_k\}} \quad \sum_{k=1}^{K} \text{tr}(\mathbf{W}_k) + \text{tr}(\mathbf{Z}) \tag{P3}$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \tag{20}$$

$$\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \tag{21}$$

$$\text{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track},p}, \quad \forall p \tag{22}$$

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \tag{23}$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \tag{24}$$

$$\mathbf{Z} \succeq \mathbf{0} \tag{25}$$

$$\mu_k \geq 0, \quad \forall k \tag{26}$$

where $\mathbf{E}_m$ is the selection matrix for AP m, and $\mu_k \geq 0$ is the S-Procedure slack variable.

Table 2: Problem (P3) Constraint Structure and Variable Mapping

Constraint	Type	Variables	Physical Meaning
(20)	LMI ($MN_t + 1$ dim.)	$\mathbf{W}_k, \mu_k$	Communication worst-case robustness
(21)	Linear inequality	$\mathbf{Z}$	Sensing detection probability
(22)	Linear inequality	$\mathbf{R}_X$	Tracking accuracy (PCRB)
(23)	Linear inequality	$\mathbf{R}_X$	Per-AP peak power
(24)–(25)	PSD cone	$\mathbf{W}_k, \mathbf{Z}$	SDR relaxation
(26)	Non-negative orthant	μ_k	S-Procedure slack

Remark 3 (Why This Is the "Solution to the Problem")**.** *Notice that every line in the above model: the objective function is a linear trace, (20) is a standard LMI, (21)–(23) are linear affine inequalities, and (24)–(26) are positive semidefinite cone constraints. There are no fractions, no quadratic forms, and no discrete variables. This mathematical form is the most standard paradigm in convex optimization. One simply declares* `variable W(N,N) semidefinite`*, calls CVX with the MOSEK solver, and the engine returns the globally optimal covariance matrix in polynomial time. Meanwhile, (20) covers worst-case*

communication robustness, (21) guarantees sensing detection probability, (22) satisfies tracking accuracy, and (23) constrains distributed AP peak power—full coverage of physical constraints.

5.3 Tightness Conditions

Theorem 1 (SDR Tightness)**.** *For the total power minimization problem, when $K \leq 2$, SDR is tight, i.e., the optimal solution satisfies* $\text{rank}(\mathbf{W}_k^*) = 1$.

Remark 4. *In the high-SNR regime, SDR is approximately tight with controllable performance loss. For the general case, beam recovery via Gaussian randomization provides a performance loss upper bounded by $O(1/L)$ (where L is the number of candidates).*

6 S-Procedure Robust Constraint Transformation

6.1 Communication Robust Constraint (Exact Transformation)

The worst-case SINR constraint is:

$$\min_{\|\Delta \mathbf{h}_k\|_2 \leq \epsilon_h} \frac{|(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_k|^2}{\sum_{j \neq k} |(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_j|^2 + \sigma_c^2} \geq \gamma_k \tag{27}$$

Using the S-Procedure, this infinite-dimensional constraint can be transformed into a finite-dimensional LMI. Define:

$$\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j \tag{28}$$

Then there exists $\mu_k \geq 0$ such that the following LMI holds:

$$\begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0} \tag{18}$$

Proposition 1. *The S-Procedure is an **exactly equivalent** transformation, not an approximation. That is, the original worst-case constraint and the LMI constraint are completely equivalent.*

6.2 Sensing Robust Constraint (Optional)

Similarly, for the sensing channel, the LMI form is:

$$\begin{bmatrix} \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z} + \nu_p \mathbf{I} & \frac{1}{\gamma_S^{\text{PoD}}}\mathbf{Z}\hat{\mathbf{g}}_p \\ \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z} & \frac{1}{\gamma_S^{\text{PoD}}}\hat{\mathbf{g}}_p^H \mathbf{Z}\hat{\mathbf{g}}_p - \sigma_s^2 - \nu_p \epsilon_g^2 \end{bmatrix} \succeq \mathbf{0} \tag{19}$$

where $\nu_p \geq 0$ is the sensing S-Procedure slack variable.

Remark 5 (LMI Typesetting Optimization)**.** *The matrix elements in (19) contain denominators γ_S^{PoD}. In standard LMI notation, both sides of the inequality can be multiplied by γ_S^{PoD}, yielding $\mathbf{Z} + \nu_p \gamma_S^{PoD} \mathbf{I}$ in the upper-left corner, thereby avoiding fractional structures inside the matrix for more elegant typesetting. Mathematical equivalence is preserved.*

Assumption 2 (Sensing Channel Perfection)**.** *This work focuses on robust design for the communication link, assuming that sensing channels are self-calibrated through tracking echoes within the current time slot with negligible error. Specifically:*

1. *Target state parameters $\boldsymbol{\theta}_p$ are accurately predicted within a short time slot, with prediction errors much smaller than one wavelength*

2. *Sensing channels $\mathbf{g}_p$ are calibrated through tracking echoes from the previous time slot, with errors incorporated into the next slot update*

3. *Sensing tasks adopt a "detection-tracking" cascade architecture: conservative thresholds during detection, and time-slot smoothness compensates for errors during tracking*

Remark 6. *Sensing channels are self-calibrating (the probing signal is transmitted and the echo is received, with the echo carrying current channel state), whereas communication channels rely on uplink pilot estimation, where pilot contamination leads to error accumulation. In top-tier journal system modeling, focusing on communication-side pilot contamination while assuming the radar side benefits from LOS echoes and Kalman tracking filtering for precise prediction is a common and prudent compromise.*

7 Proof of Sensing Constraint Convexity

7.1 PCRB Constraint Convexity

Constraint: $\mathrm{tr}(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}$

- $\mathbf{F}_p$ is a known constant positive semidefinite matrix
- $\mathbf{R}_X$ is the optimization variable (positive semidefinite matrix)
- $\mathrm{tr}(\mathbf{F}_p\mathbf{R}_X)$ is a linear function of $\mathbf{R}_X$

Conclusion: Linear inequality constraint, naturally convex.

7.2 Sensing SINR Constraint Convexity

Constraint: $\mathrm{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}}\sigma_s^2$

- $\mathbf{g}_p\mathbf{g}_p^H$ is a known constant positive semidefinite matrix
- $\mathbf{Z}$ is the optimization variable
- $\mathrm{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z})$ is a linear function of $\mathbf{Z}$

Conclusion: Linear inequality constraint, naturally convex.

7.3 Power Constraint Convexity

Constraint: $\mathrm{tr}(\mathbf{E}_m\mathbf{R}_X) \leq P_{\max}$

- $\mathbf{E}_m$ is a known constant matrix
- $\mathbf{R}_X$ is the optimization variable

Conclusion: Linear inequality constraint, naturally convex.

7.4 Summary

Table 3 summarizes the convexity of each constraint.

Table 3: Convexity Summary of Constraints

Constraint	Form	Convexity
Communication Robust (S-Procedure)	LMI	Convex
PCRB	Linear trace	Convex
Sensing SINR	Linear trace	Convex
Power	Linear trace	Convex
Positive Semidefinite	$\mathbf{W}_k, \mathbf{Z} \succeq \mathbf{0}$	Convex

Theorem 2 (Final Problem Convexity)**.** *Problem (P3) is a standard convex SDP, solvable to arbitrary precision via interior-point methods in polynomial time. Combined with Algorithm 22 (Gaussian randomization rank-one recovery), this theoretical framework can be directly translated into MATLAB/CVX code.*

8 Rank-One Recovery and Fallback Scheme

8.1 Problem Background

SDR relaxation discards the $\text{rank}(\mathbf{W}_k) = 1$ constraint. After solving:

- If $\text{rank}(\mathbf{W}_k^*) = 1$: directly recover $\mathbf{w}_k$ through eigenvalue decomposition
- If $\text{rank}(\mathbf{W}_k^*) > 1$: Gaussian randomization is required to extract a suboptimal beam

8.2 Algorithm 1: Gaussian Randomization Rank-One Recovery

The following pseudocode describes the rank-one recovery algorithm. The algorithm first checks the rank of the communication covariance matrix; if the rank is 1, the beam is directly extracted. If the rank exceeds 1, candidate beams are generated via Gaussian randomization, and the best candidate satisfying the constraints is selected. If still unsatisfied, power scaling is employed as a fallback.

8.3 Power Scaling Mathematical Guarantee

After power scaling:

- Per-AP power: $P'_m = \beta_m P_m = P_{\max}$ (strictly satisfied)
- Communication SINR: $\text{SINR}'_k = \beta_m \cdot \text{SINR}_k$ (linear scaling)
- Sensing SINR: $\text{SINR}'_{S,p} = \beta_m \cdot \text{SINR}_{S,p}$ (linear scaling)

Remark 7. *The rank of the sensing covariance $\mathbf{Z}^*$ exceeding 1 is a* ***design intention*** *(multi-target coverage), not a recovery failure. In pure communication MU-MIMO, power minimization tends to form "pencil beams" (rank-one); in ISAC, sensing tasks require spatial energy dispersion, and multi-rank is a design intention. The "role separation" between communication and sensing in the covariance domain is the essential characteristic of ISAC waveform design.*

Algorithm 1: Gaussian Randomization Rank-One Recovery

Input: SDP optimal solution $\{\mathbf{W}_k^*\}_{k=1}^K$, $\mathbf{Z}^*$, constraint parameters
Output: Beams $\{\mathbf{w}_k\}_{k=1}^K$ satisfying all constraints, sensing waveform

`// Step 1: Communication beam recovery`

foreach *User* k **do**
  if $\text{rank}(\mathbf{W}_k^*) = 1$ **then**
    $\mathbf{w}_k \leftarrow \sqrt{\lambda_{\max}(\mathbf{W}_k^*)} \cdot \mathbf{v}_{\max}(\mathbf{W}_k^*)$; `// Direct extraction`
  else
    Generate $L = 1000$ candidates: $\boldsymbol{\xi}_l \sim \mathcal{CN}(\mathbf{0}, \mathbf{W}_k^*)$;
    Normalize: $\mathbf{w}_k^{(l)} \leftarrow \sqrt{\text{tr}(\mathbf{W}_k^*)} \cdot \frac{\boldsymbol{\xi}_l}{\|\boldsymbol{\xi}_l\|_2}$;
    Compute constraint violation:
    $v_l \leftarrow \sum_{k'} \max(0, \gamma_k - \text{SINR}_{k'}^{(l)}) + \sum_p \max(0, \gamma_S^{\text{PoD}} - \text{SINR}_{S,p}^{(l)})$;
    $l^* \leftarrow \arg\min_l v_l$;
    if $v_{l^*} = 0$ **then**
      Accept $\mathbf{w}_k \leftarrow \mathbf{w}_k^{(l^*)}$;
    else
      Enter power scaling fallback (Step 3);

`// Step 2: Sensing waveform recovery`

Eigenvalue decomposition: $\mathbf{Z}^* = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^H$;
Generate multi-stream transmission waveform: $\mathbf{z}_p = \sum_{i=1}^r \sqrt{\lambda_i} \mathbf{v}_i s_i$, where $s_i \sim \mathcal{CN}(0, 1)$ are independent;

`// Step 3: Power scaling fallback`

foreach *AP* m **do**
  $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m)$;
  if $P_m > P_{\max}$ **then**
    $\beta_m \leftarrow P_{\max}/P_m$;
    $\mathbf{w}_{m,k} \leftarrow \mathbf{w}_{m,k} \cdot \sqrt{\beta_m}$;
    $\mathbf{Z}_m \leftarrow \mathbf{Z}_m \cdot \beta_m$;

`// Step 4: Performance guarantee`

if $K \leq 2$ **then** SDR is tight, optimal solution recovered;
if $K > 2$ **then** Gaussian randomization provides suboptimal solution with performance loss upper bounded by $O(1/L)$;

9 Closed-Form Solution Derivation

9.1 Assumptions

To obtain closed-form solutions, the problem is simplified under specific conditions:

- Ignore PCRB constraints (or assume they are automatically satisfied)
- Use ZF communication beams
- Use matched-filter sensing beams
- Assume relaxed per-AP power constraints

9.2 ZF Beamforming

Condition: $\mathrm{rank}(\mathbf{H}_{\text{all}}) \geq K$, i.e., $M^{\text{all}} N_t \geq K$.

The ZF matrix is:

$$\mathbf{W}_{\text{ZF}} = \mathbf{H}_{\text{all}} (\mathbf{H}_{\text{all}}^H \mathbf{H}_{\text{all}})^{-1} \tag{20}$$

Satisfying the orthogonality property:

$$\mathbf{h}_k^{\text{all},H} \mathbf{W}_{\text{ZF}}(:, j) = \delta_{kj} \tag{29}$$

Power allocation:

$$p_k = \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:, k)\|_2^2 \tag{21}$$

9.3 Sensing Beams

$$\mathbf{z}_p = \sqrt{P_{S,p}} \frac{\mathbf{g}_p^{\text{all}}}{\|\mathbf{g}_p^{\text{all}}\|_2} \tag{22}$$

where $P_{S,p} = (\gamma_S^{\text{PoD}})^{\text{robust}} \sigma_s^2 / \|\mathbf{g}_p^{\text{all}}\|_2^2$.

9.4 Algorithm 2: Closed-Form Solver

The following pseudocode describes the closed-form solver. The algorithm first selects the optimal AP subset for each target based on sensing channel strength, then computes ZF communication beams and matched-filter sensing beams on the selected APs, and finally verifies power constraints before returning a feasible solution.

10 Complexity Analysis

10.1 SDP Solving Complexity

Number of variables:

- K matrices $\mathbf{W}_k$: $K \cdot (MN_t)^2$ real variables
- 1 matrix $\mathbf{Z}$: $(MN_t)^2$ real variables
- K variables μ_k: K real variables

Algorithm 2: Cell-Free ISAC Closed-Form Solver

Input: $\mathbf{H}, \mathbf{G}, M, N_t, K, P, P_{\max}, \{\gamma_k\}, \gamma_S^{\text{PoD}}, N_{\text{req}}, \epsilon_h, \epsilon_g$

Output: $\{\mathbf{w}_{m,k}\}, \{\mathbf{Z}_m\}, \{b_{mp}\}, P_{\text{total}}$

`// Step 1: Compute robust thresholds`

$\gamma_k^{\text{robust}} \leftarrow \gamma_k \cdot (1+\epsilon_h)^2/(1-\epsilon_h)^2$;

$\gamma_S^{\text{robust}} \leftarrow \gamma_S^{\text{PoD}} \cdot (1+\epsilon_g)^2/(1-\epsilon_g)^2$;

`// Step 2: AP selection`

for $p = 1, \dots, P$ **do**

  Compute $\|\mathbf{g}_{m,p}\|_2$ for all m;

  Select top-N_{req} APs: $b_{mp} = 1$;

$\mathcal{M}^{\text{all}} \leftarrow \bigcup_p \{m : b_{mp} = 1\}$;

`// Step 3: Extract sub-channels`

$\mathbf{H}^{\text{all}} \leftarrow \mathbf{H}(\mathcal{M}^{\text{all}}, :)$;

$\mathbf{g}_p^{\text{all}} \leftarrow \mathbf{G}(\mathcal{M}^{\text{all}}, p)$ for all p;

`// Step 4: Communication beams`

if $\text{rank}(\mathbf{H}^{all}) \geq K$ **then**

  $\mathbf{W}_{\text{ZF}} \leftarrow \mathbf{H}^{\text{all}} \cdot (\mathbf{H}^{\text{all},H}\mathbf{H}^{\text{all}})^{-1}$;

  for $k = 1, \dots, K$ **do**

    $\mathbf{w}_k^{\text{ZF}} \leftarrow \mathbf{W}_{\text{ZF}}(:, k)/\|\mathbf{W}_{\text{ZF}}(:, k)\|_2$;

    $p_k \leftarrow \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:, k)\|_2^2$;

    $\mathbf{w}_k \leftarrow \sqrt{p_k} \cdot \mathbf{w}_k^{\text{ZF}}$;

else

  Use MRT (suboptimal);

`// Step 5: Sensing beams`

for $p = 1, \dots, P$ **do**

  $P_{S,p} \leftarrow \gamma_S^{\text{robust}} \sigma_s^2 / \|\mathbf{g}_p^{\text{all}}\|_2^2$;

  $\mathbf{z}_p \leftarrow \sqrt{P_{S,p}} \cdot \mathbf{g}_p^{\text{all}} / \|\mathbf{g}_p^{\text{all}}\|_2$;

  $\mathbf{Z}_m \leftarrow \mathbf{z}_p \mathbf{z}_p^H$ (rank-1);

`// Step 6: Power check`

for $m \in \mathcal{M}^{all}$ **do**

  $P_m \leftarrow \sum_k \|\mathbf{w}_{m,k}\|_2^2 + \text{tr}(\mathbf{Z}_m)$;

  if $P_m > P_{\max}$ **then**

    Scale or mark infeasible;

`// Step 7: Verify and return`

- Total: $O(K(MN_t)^2)$

Number of constraints:

- K communication LMIs: $K \cdot (MN_t + 1) \times (MN_t + 1)$
- P sensing linear constraints
- M power constraints
- Total computational complexity: $O(K(MN_t)^3)$

Solving time: $O((MN_t)^{3.5})$ — for $M = 16, N_t = 4$, approximately 5–10 seconds (MOSEK).

10.2 Closed-Form Solution Complexity

Table 4 lists the complexity of each step in the closed-form solver.

Table 4: Complexity Analysis of Closed-Form Solver

Step	Operation	Complexity
AP Selection	Sorting per target	$O(MP \log M)$
ZF Inversion	$(\mathbf{H}^H\mathbf{H})^{-1}$	$O(K^3)$
Communication Power	K multiplications	$O(K)$
Sensing Beams	P normalizations	$O(PMN_t)$
Power Check	M summations	$O(MK)$
Total		$O(MP \log M + K^3)$

For $M = 16, N_t = 4, K = 10, P = 4$: $O(7400)$, i.e., $less than 0.1$ second.

11 Feasibility Conditions

11.1 Necessary Conditions

ZF feasibility:

$$M^{\text{all}} N_t \geq K \tag{30}$$

For $N_t = 4, K = 10$: $M^{\text{all}} \geq 3$ (theoretical), $M^{\text{all}} \geq 4$ (numerically stable).
Power feasibility:

$$P^{\min}_{\text{comm}} + P^{\min}_{\text{sens}} \leq M \cdot P_{\max} \tag{31}$$

where:

$$P^{\min}_{\text{comm}} = \sum_{k=1}^{K} \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:,k)\|_2^2 \tag{32}$$

$$P^{\min}_{\text{sens}} = \sum_{p=1}^{P} (\gamma_S^{\text{PoD}})^{\text{robust}} \frac{\sigma_s^2}{\|\mathbf{g}_p^{\text{all}}\|_2^2} \tag{33}$$

11.2 Infeasible Scenarios

1. Target is far from all APs ($d > 50$ m)
2. Excessive number of users ($K > M^{\text{all}} N_t$)
3. Insufficient power budget ($P_{\max} < P_{\text{comm}}^{\min}$)
4. Excessive CSI error ($\epsilon > 0.5$)

12 Key Formulas Summary

Table 5 summarizes the core formulas of this paper.

Table 5: Summary of Key Formulas

Number	Name	Expression
(3)	Communication robustness factor	$\eta_h = \left(\frac{1-\epsilon_h}{1+\epsilon_h}\right)^2$
(4)	Communication robust threshold	$\gamma_k^{\text{robust}} = \gamma_k / \eta_h$
(9)	Sensing robustness factor	$\eta_g = \left(\frac{1-\epsilon_g}{1+\epsilon_g}\right)^2$
(10)	Sensing robust threshold	$(\gamma_S^{\text{PoD}})^{\text{robust}} = \gamma_S^{\text{PoD}} / \eta_g$
(12)	FIM trace constraint	$\text{tr}(\mathbf{J}_p^{\text{data}}) = \text{tr}(\mathbf{F}_p \mathbf{R}_X)$
(13)	FIM constant matrix	$\mathbf{F}_p = \frac{2}{\sigma_s^2} \text{Re}\left\{\nabla_{\boldsymbol{\theta}_p}^H \mathbf{g}_p \nabla_{\boldsymbol{\theta}_p} \mathbf{g}_p^H\right\}$
(18)	Communication S-Procedure LMI	$\begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}$
(19)	Sensing S-Procedure LMI (optional)	$\begin{bmatrix} \frac{1}{\gamma_S^{\text{PoD}}} \mathbf{Z} + \nu_p \mathbf{I} & \frac{1}{\gamma_S^{\text{PoD}}} \mathbf{Z} \hat{\mathbf{g}}_p \\ \frac{1}{\gamma_S^{\text{PoD}}} \hat{\mathbf{g}}_p^H \mathbf{Z} & \frac{1}{\gamma_S^{\text{PoD}}} \hat{\mathbf{g}}_p^H \mathbf{Z} \hat{\mathbf{g}}_p - \sigma_s^2 - \nu_p \epsilon_g^2 \end{bmatrix} \succeq \mathbf{0}$
(20)	ZF matrix	$\mathbf{W}_{\text{ZF}} = \mathbf{H}(\mathbf{H}^H \mathbf{H})^{-1}$
(21)	Communication power allocation	$p_k = \gamma_k^{\text{robust}} \sigma_c^2 \|\mathbf{W}_{\text{ZF}}(:, k)\|_2^2$

13 Conclusion and Future Work

This paper establishes a complete mathematical framework for joint beamforming optimization in Cell-Free ISAC systems. The main contributions include:

1. **Problem Formulation**: Established the standard optimization problem (P1) incorporating communication QoS, sensing detection, tracking accuracy, and AP selection constraints.
2. **Robustness Analysis**: Derived worst-case SINR constraints under CSI errors, introducing robustness factors η_h and η_g to quantify the impact of estimation errors.
3. **Convex Relaxation**: Transformed the non-convex rank-one constraint into a semidefinite constraint via SDR, and proved the convexity of sensing constraints under the single-angle estimation assumption.

4. **S-Procedure Transformation**: Exactly transformed infinite-dimensional worst-case constraints into finite-dimensional LMIs, guaranteeing mathematical equivalence.

5. **Rank-One Recovery**: Proposed Gaussian randomization algorithms with power scaling fallback schemes to ensure solution feasibility.

6. **Closed-Form Solution**: Provided ZF beams with matched-filter sensing under simplified conditions, with complexity $O(MP \log M + K^3)$.

Future work:

- Implement the SDP solver (requires MOSEK support for LMI constraints)
- Extend sensing channel robustness (Options B/C)
- Multi-target joint optimization verification
- Simulation validation and performance evaluation